Sodium

Sodium is the principal cation of extracellular fluid. Human body has approximately 1.8 g of sodium per kilogram of body weight. The body regulates the sodium concentration in the extracellular fluid within narrow limits.

Function

In addition to its role in maintaining the body's water balance, sodium functions in maintaining extracellular fluid volume and in regulating the body's acid – base balance.

Food sources

The major dietary source of sodium is sodium chloride, more commonly called table salt. One teaspoon of table salt supplies 2132 milligrams of sodium.

Deficiency

Excessive sodium loss, not a low dietary intake; is the usual cause of sodium deficiency. Trauma, long – term diarrhea, vomiting and kidney disease may also lead to excessive sodium losses.

A sodium deficiency, with fluid levels remaining constant or increasing, leads to a decrease in the extracellular sodium concentration. As a result, water migrates into cells, leading to water intoxication. Water intoxication causes loss of appetite, muscle twitching, mental apathy, coma, and seizures.

Potassium

The electrolyte potassium is found mainly in the intracellular fluid

Functions

- The main function of potassium, like sodium, is to maintain water balance.
- Potassium is required for maintaining a normal heartbeat.

Food sources

This mineral is widely distributed in foods, but fruits and vegetables are generally the most nutrient – dense sources of potassium.

Deficiency

As with sodium, deficiencies of potassium are usually caused by excessive losses, not low intakes. Losses occur primarily via the urine; lesser amounts are lost in sweat. Excessive potassium losses via the kidneys may result from use of potassium – depleting diuretics, which are often prescribed to treat high blood pressure. High levels of activity in hot, humid climates may lead to excessive potassium losses via perspiration. Much potassium can also be lost through long-term vomiting or prolonged diarrhea.

Symptoms of a potassium deficiency include weakness, loss of appetite, nausea, listlessness, apprehension, fatigue, irrational behavior, muscle weakness, and muscle cramping. A severe deficiency may cause an abnormal heartbeat and possibly death.

Chloride

The electrolyte chloride is found primarily in the extracellular fluid.

Functions

- Chloride's major functions are maintaining the body's water and electrolyte balance.
- Chloride is mainly a component of hydrochloric acid, which is secreted in the stomach and helps in the digestion of protein.

Food sources

The main dietary sources of chloride is sodium chloride, that is, table salt. Chloride is also found in many processed food that contain added salt.

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